BPBIO320 Series / BPBIO750 Communication Protocol

Approver: Jongheun Shin

Checker: Daesuk Kim

Author: Jeonghyeon Son

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Rev.	Detail	Date	Author
V1.0	Initial Version	2020.04.03	Kyunghoon Jung
V1.1	Function Added.(Need F/W Check)	2021.10.25	Jongheun Shin
V1.2	CTS Protocol, MID Protocol IR Sensor, Elbow Sensor detect.	2022.07.12	Jongheun Shin
V1.3	Documentation improvements	2026.08.11	Jeonghyeon Son

1 Basic protocol frame structure

- Protocol Frame

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0x16	0x16	0x01	0	0	0x02	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0x1E	R	B	0x1E	0	0	0	0	0
0x1E	**S**	**char data**	**char data**	**char data**	0x1E	M	0	0	0	0x1E	**D**	**char data**	**char data**	**char data**	0x1E
P	**char data**	**char data**	**char data**	0x1E	0	0	0	0	0	0	0	0	0x1E	0x03	BCC

** Only the fields highlighted in yellow are meaningful. **All other fields must be ignored** — do not validate them. The values shown for those fields are placeholders only and are not guaranteed; the actual transmitted values may differ.

** The information is transmitted by using the position of values with yellow mark.

example) Systolic - 120mmHg, Diastolic - 80mmHg, Pulse Rate - 75bpm

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0x16	0x16	0x01	0	0	0x02	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0x1E	R	B	0x1E	0	0	0	0	0
0x1E	**S**	**1**	**2**	**0**	0x1E	M	0	0	0	0x1E	**D**	**0**	**8**	**0**	0x1E
P	**0**	**7**	**5**	0x1E	0	0	0	0	0	0	0	0	0x1E	0x03	BCC

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Open Protocol

2. BPBIO320 Series / BPBIO750 ➔ APP

(Automatic transmission according to device operating status)

1) When measurement completed, result data transferred to APP

S	1	2	0
D	0	8	0
P	0	7	5

example) Systolic - 120mmHg, Diastolic - 80mmHg, Pulse Rate - 75bpm

2) When measurement starts

S	S	T	A
D	S	T	A
P	S	T	A

3) When measurement interrupted by pressing start/stop button of BP monitor

S	S	T	P
D	S	T	P
P	S	T	P

4) When measurement interrupted by errors of BP monitor

S	E	r	r
D	0	0	1
P	E	r	r

example) Error number - 001

5) When device has detect user. (BPBIO320S, BPBIO750)

S	A	W	K
D	A	W	K
P	A	W	K

V1.2 Added (MID mode only)

6) Elbow detected

S	E	L	B
D	E	L	B

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P	E	L	B

7) Elbow removed

S	e	l	b
D	e	l	b
P	e	l	b

3. APP ➔ BPBIO320 Series / BPBIO750: Request recent measurement result

- Frame

SYN	SYN	SOH			STX	Command	ETC	BCC
0x16	0x16	0x01	0x30	0x30	0x02	0x52 0x42	0x03	0x10

- Response

S	1	2	0
D	0	8	0
P	0	7	5

4. APP ➔ BPBIO320 Series / BPBIO750: Start measurements command

- Frame

SYN	SYN	SOH			STX	Command	ETC	BCC
0x16	0x16	0x01	0x30	0x30	0x02	0x52 0x43	0x03	0x11

- Response

The BP monitor starts to measure.

5. APP ➔ BPBIO320 Series / BPBIO750: Stop measurements command

- Frame

SYN	SYN	SOH			STX	Command	ETC	BCC
0x16	0x16	0x01	0x30	0x30	0x02	0x52 0x46	0x03	0x14

- Response

The BP monitor stop to measure.

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6. APP ➔ BPBIO320 Series / BPBIO750: Confirm communication connection

- Frame

SYN	SYN	SOH			STX	Command	ETC	BCC
0x16	0x16	0x01	0x30	0x30	0x02	0x52 0x45	0x03	0x17

- Response

S	C	O	N
D	C	O	N
P	C	O	N

7. APP ➔ BPBIO320 Series / BPBIO750: Start/Stop button will lock

- Frame

SYN	SYN	SOH			STX	Command	ETC	BCC
0x16	0x16	0x01	0x30	0x30	0x02	0x52 0x4C	0x03	0x1E

- Response

S	L	C	K
D	L	C	K
P	L	C	K

8. APP ➔ BPBIO320 Series / BPBIO750: Start/Stop button will release

- Frame

SYN	SYN	SOH			STX	Command	ETC	BCC
0x16	0x16	0x01	0x30	0x30	0x02	0x52 0x72	0x03	0x20

- Response

S	R	L	S
D	R	L	S
P	R	L	S

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V1.2 Added

9. BPBIO320/750 ➔ APP, Real time pressure (MID mode only)

- Response

STX	ID	BOD0	BOD1	CMD0	CMD1	Data	CheckSum	ETX
0x02	'B'	?	?	'C'	'P'	Pressure	?	0x03

- 35mmHg -> '035' + 0x1b

10. APP ➔ BPBIO320/750, Request envelope Data

- Frame

STX	ID	BOD0	BOD1	CMD0	CMD1	Data	CheckSum	ETX
0x02	'B'	?	?	'C'	'E'	Empty	?	0x03

- Response

STX	ID	BOD0	BOD1	CMD0	CMD1	Data	CheckSum	ETX
0x02	'B'	?	?	'C'	'E'	Data	?	0x03

- Data Frame (0x1b: Delimiter)

Offset	Length	Contents
0	1byte	'S', opening identifier of pressure graph
1	1byte	Hundreds digit of pressure value
2	1byte	Tens digit of pressure value
3	1byte	Units digit of pressure value
4	1byte	0x1b
5	1byte	'E', end delimiter of pressure graph
6	1byte	Hundreds digit of pressure value
7	1byte	Tens digit of pressure value
8	1byte	Units digit of pressure value
9	1byte	0x1b
10	1byte	'D', Envelope Data delimiter
11	1byte	First Pulse graph Data_ Hundreds digit of pressure value
12	1byte	First Pulse graph Data_ Tens digit of pressure value
13	1byte	First Pulse graph Data_ Units digit of pressure value

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14	1byte	0x1b
15	1byte	Second Pulse graph Data_ Hundreds digit of pressure value
16	1byte	Second Pulse graph Data_ Tens digit of pressure value
17	1byte	Second Pulse graph Data_ Units digit of pressure value
18	1byte	0x1b
…	1byte	…
167	1byte	40th Pulse graph Data_ Hundreds digit of pressure value
168	1byte	40th Pulse graph Data_ Tens digit of pressure value
169	1byte	40th Pulse graph Data_ Units digit of pressure value
170	1byte	0x1b

11. APP ➔ BPBIO320/750, Request latest result

- Frame

STX	ID	BOD0	BOD1	CMD0	CMD1	Data	CheckSum	ETX
0x02	'B'	?	?	'C'	'R'	Empty	?	0x03

- Response

STX	ID	BOD0	BOD1	CMD0	CMD1	Data	CheckSum	ETX
0x02	'B'	?	?	'C'	'R'	Resut	?	0x03

- Result : SYS, DIA, PUL, Arm Sensing, Irregular Pulse

Ex) '120' + 0x1b + '080' + 0x1b + '070' + 0x1b + 'AX' + 0x1b + 'IX' + 0x1b

ASCII Code 12 Byte, 0x1b: Delimiter

Elbow Status(A) ~ X = '1'(Removed) or '0'(Detected)

Irregular Pulse(I) ~ X = '1'(Detected) or '0'(Not Detected)

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